

Group Comparisons and Statistical Modelling with R

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i Chapter overview

Chapter 5 made estimates, intervals and tests executable in base R. Here we return to Chapter 4's group comparisons and use R to connect a fitted model with the question it answers. We begin with three condition means from Chapter 4's background-sound experiment and compare selected differences inside stated families. We then examine a six-cell pattern from its feedback-by-practice study and test how one comparison changes across a second factor by comparing two named models on the same records.

Counts require a different response model, and the size of a difference between two groups' proportions can be stated on several scales, which we set side by side. The same counts provide a small example of likelihood, which makes a fitting criterion explicit. Finally, simulation examines procedures under a stated generating process, and planning relates sample size to either power or interval width. Throughout, a useful result keeps its data, target, uncertainty calculation and conditions together.

Studies rarely stop at one mean. Chapter 4's invented background-sound experiment compares mean reading comprehension under three conditions, silence, instrumental music and music with lyrics, and its second invented study asks how the difference between spaced and massed practice changes across three kinds of feedback. Means that differ in a record differ for more than one reason, so the question is what a stated model and its conditions let us say about several means at once, and what a single answer to that question leaves unsettled. Both studies are invented teaching inputs, so each analysis shows what a procedure reports, not what background sound or feedback does.

1 Fit and read group means

1.1 Begin with the question and records

Does mean reading comprehension differ across three background-sound conditions? Chapter 4 developed this question with an invented experiment. Each of thirty first-year university students read the same passage for 20 minutes and then answered a 20-item reading-comprehension quiz, scored from 0 to 20. Ten students were randomly assigned to each sound condition: Silence, Instrumental music or Music with lyrics. Each student supplies one quiz score, so Table 1 holds thirty scores from thirty different students.

Table 1

The same 30 invented comprehension scores as Chapter 4.

Sound condition	Scores (out of 20)
Silence	8, 9, 9, 11, 12, 13, 13, 14, 15, 16
Instrumental music	6, 8, 8, 9, 9, 11, 11, 12, 12, 14
Music with lyrics	4, 5, 6, 7, 8, 8, 9, 10, 11, 12

These values were constructed to keep the calculations transparent. They are teaching inputs, not empirical findings about music and reading. The student is both the unit of random assignment and the unit of measurement: the ten scores in a condition come from ten different students, not from repeated measurements of one student. Silence is the natural reference condition, so it comes first.

The following code creates the complete data. The factor levels fix the display order, make Silence the reference level for later model coefficients and fix the direction of each subtraction. The course data file `background_sound.csv` holds the same thirty rows with a student identifier; reading it with `read.csv()` and applying the same `factor()` levels gives identical results.

```
sound_data <- data.frame(
  sound = factor(rep(c("Silence", "Instrumental music",
    "Music with lyrics"), each = 10),
  levels = c("Silence", "Instrumental music",
    "Music with lyrics")),
  score = c(8, 9, 9, 11, 12, 13, 13, 14, 15, 16,
    6, 8, 8, 9, 9, 11, 11, 12, 12, 14,
    4, 5, 6, 7, 8, 8, 9, 10, 11, 12))
table(sound_data$sound)
#>
#>          Silence Instrumental music  Music with lyrics
#>              10              10              10
```

```
with(sound_data, tapply(score, sound, mean))
#>      Silence Instrumental music Music with lyrics
#>          12             10             8
with(sound_data, tapply(score, sound, sd))
#>      Silence Instrumental music Music with lyrics
#>  2.708012802    2.403700850    2.581988897
```

Each condition holds ten students. The condition means are 12, 10 and 8 correct answers, and the grand mean of all thirty scores is 10. The SDs, 2.7080, 2.4037 and 2.5820, are similar without being identical. Before asking whether a difference is large relative to its uncertainty, we can see each difference on the quiz's own scale: Instrumental music lies 2 points and Music with lyrics 4 points below Silence.

Figure 1 shows every student beside the three condition means. Each point is one student's score, moved a small random distance sideways so that equal scores in a condition stay visible, and each diamond marks a condition mean. Two kinds of variation appear together. The diamonds step down by 2 points from Silence to Instrumental music and by another 2 points to Music with lyrics, while the ten scores in each condition spread across 8 points around their own diamond. Whether a 2-point step is large depends on that spread within conditions, and the model below turns this visual comparison into numbers.

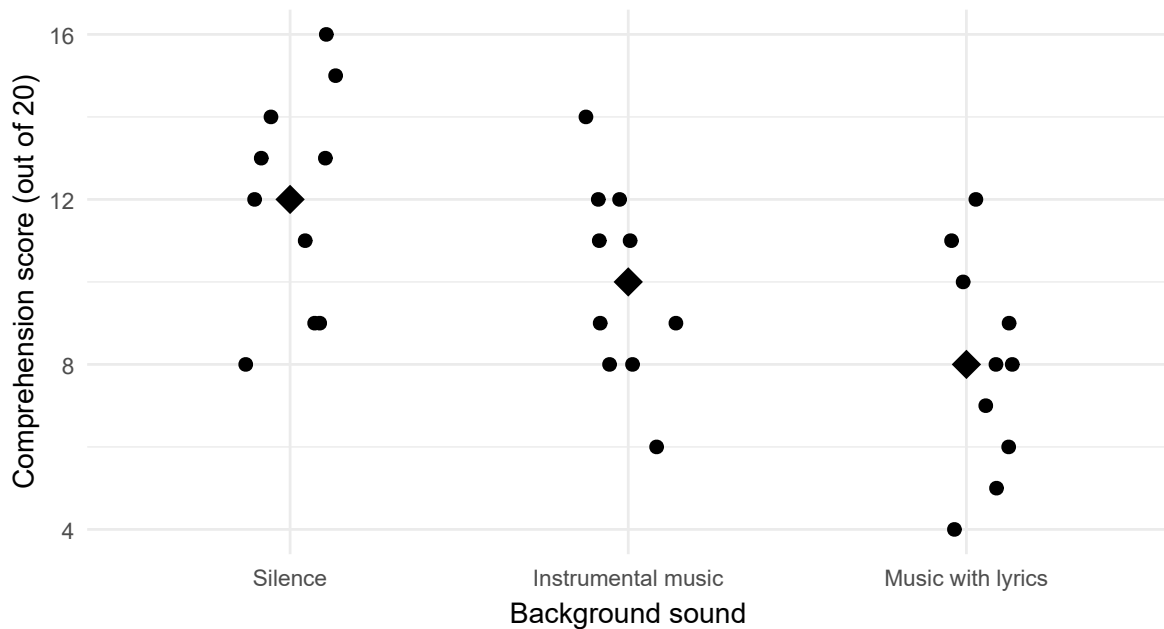


Figure 1: Individual comprehension scores and their three condition means.

The figure is drawn with the `ggplot2` package, which describes a graph as a data frame, a mapping of variables to position or colour, and layers of marks; Computer Lab 6 builds it one layer at a time and sets this way of working beside base R and the wider tidyverse.

1.2 Fit the group model and read its decomposition

For student i in sound condition j , the one-way model is

$$Y_{ij} = \mu_j + \varepsilon_{ij}.$$

Here μ_j is the condition's process mean and ε_{ij} is its student's deviation from that mean. Fitting by least squares estimates each μ_j with the corresponding sample mean, so $\widehat{Y}_{ij} = \bar{Y}_j$. The residual is $e_{ij} = y_{ij} - \bar{y}_j$: an observed score minus its fitted condition mean. For Silence the fitted values are all 12 and the residuals are $-4, -3, -3, -1, 0, 1, 1, 2, 3, 4$.

The model separates differences between conditions from differences among students within a condition. Chapter 4's decomposition becomes

$$\begin{aligned} SS_T &= \sum_j \sum_i (y_{ij} - \bar{y})^2 = SS_B + SS_W, \\ SS_B &= \sum_j n_j (\bar{y}_j - \bar{y})^2, \\ SS_W &= \sum_j \sum_i (y_{ij} - \bar{y}_j)^2. \end{aligned}$$

The total measures deviation from the grand mean. The between-group term measures deviation of the three condition means from that grand mean, weighted by their group counts. The within-group term measures deviation of students from their own fitted condition means. In this example, $SS_B = 10(12 - 10)^2 + 10(10 - 10)^2 + 10(8 - 10)^2 = 80$. The squared residuals within Silence, Instrumental music and Music with lyrics sum to 66, 52 and 60, giving $SS_W = 178$ and $SS_T = 80 + 178 = 258$. In Figure 1 these are the steps between the diamonds and the spread of the points around them.

The formula `score ~ sound` tells `aov()` to explain score using the sound factor, and `anova()` extracts the decomposition from the fitted object.

```

one_fit <- aov(score ~ sound, data = sound_data)
anova(one_fit)
#> Analysis of Variance Table
#>
#> Response: score
#>           Df Sum Sq   Mean Sq F value    Pr(>F)
#> sound       2     80 40.000000 6.06742 0.0066651
#> Residuals  27    178  6.592593

```

Three fitted condition means use three parameters. Consequently the within-group degrees of freedom are $30 - 3 = 27$. The between-group component has $3 - 1 = 2$ degrees of freedom, and the total has $30 - 1 = 29$. Dividing each sum of squares by its degrees of freedom gives a mean square. The within-group mean square $MS_W = 178/27 = 6.5926$ estimates the common error variance.

Table 2

One-way ANOVA on the thirty background-sound scores.

Source	SS	df	MS	F	p
Sound condition	80	2	40.0000	6.0674	.0067
Within groups	178	27	6.5926		
Total	258	29			

The null hypothesis is $H_0 : \mu_S = \mu_I = \mu_L$ for Silence, Instrumental music and Music with lyrics. Under independent Normal errors with a common variance, the ratio

$$F = \frac{MS_B}{MS_W} = \frac{40}{178/27} = 6.0674$$

has an $F_{2,27}$ reference distribution when that null holds. The p-value .0067 is the probability, under this null model, of an F statistic at least this large. At a .05 threshold it provides evidence against equal process means. It does not say that every pair of conditions differs, that every kind of background sound changes comprehension, or that the invented result establishes a real effect of music on reading. The next section asks the individual comparison questions.

The observed share of score variation associated with these condition means is

$$\eta^2 = \frac{SS_B}{SS_T} = \frac{80}{258} = .3101.$$

This is an in-sample variation summary, accompanying differences expressed in score points. It is neither the probability that background sound caused a result nor a replacement for a

confidence interval. The bias-adjusted omega-squared of Chapter 4, Appendix B.2, is .2525 for these scores, and the Computer Lab rebuilds both summaries from the scores.

1.3 Express the same fit with indicators

A factor with three levels can also enter a linear model as two indicator variables. Let $D_I = 1$ for Instrumental music and 0 otherwise, and let $D_L = 1$ for Music with lyrics and 0 otherwise. Silence has both indicators 0. The fitted equation is

$$\hat{y} = 12 - 2D_I - 4D_L.$$

Substituting the indicators gives 12 for Silence, $12 - 2 = 10$ for Instrumental music and $12 - 4 = 8$ for Music with lyrics. The intercept is the reference mean; each other coefficient is a difference from that mean, negative here because both music conditions have lower means than Silence. The factor order in the original setup makes Silence the reference level.

```
indicator_fit <- lm(score ~ sound, data = sound_data)
coef(indicator_fit)
#>           (Intercept) soundInstrumental music
#>                12                -2
#> soundMusic with lyrics
#>                -4
```

`lm()` and `aov()` describe the same set of fitted means, so they give the same residuals and the same ANOVA table. The formula names the outcome and the explanatory variable; the study's design supplies the independence assumption. A different reference level changes which comparisons the coefficients name, but not the fitted means, the residuals or the table: with Instrumental music as the reference, the coefficients become 10, 2 and -2 . This indicator representation prepares the named models of Section 3.2. Chapter 7 develops a continuous predictor's slope, and Chapter 8 extends the conditional interpretation to several predictors.

2 Ask comparisons within stated families

2.1 Specify the comparison and its standard error

Suppose the two questions compare each music condition with Silence: Instrumental music versus Silence, and Music with lyrics versus Silence. In the order Silence, Instrumental music, Music with lyrics, their coefficient vectors are $(-1, 1, 0)$ and $(-1, 0, 1)$. Each vector sums to zero. A quantity

$$C = \sum_j c_j \mu_j, \quad \sum_j c_j = 0,$$

is a contrast of the process means. Its estimate replaces process means with the observed group means: $\widehat{C} = \sum_j c_j \bar{y}_j$. Thus our estimates are -2 and -4 points: both music conditions lie below Silence. Calling a comparison a contrast specifies this mathematical form; it does not say whether the comparison was planned before seeing the results or selected afterward. That is additional information about the analysis.

For independent groups with a common error variance, Chapter 4 extends Chapter 3's pooled SE, $s_p \sqrt{1/n_1 + 1/n_0}$, from two means to a contrast by replacing s_p^2 with MS_W :

$$SE(\widehat{C}) = \sqrt{MS_W \sum_j \frac{c_j^2}{n_j}}.$$

Each comparison uses two of the means, so both have estimated SE $\sqrt{(178/27)(1/10 + 1/10)} = 1.1483$ points. The third condition still contributes to the common error estimate and its 27 residual degrees of freedom. This matters when comparing these results with an analysis restricted to only two conditions.

2.2 Calibrate the two-comparison family

An individual 95% interval uses a .05 error probability for that one comparison. The declared family here is *each music condition against Silence*. To cover both of its comparisons with at least 95% familywise confidence, Bonferroni allocates .025 to each two-sided comparison. Each tail then receives $.05/(2 \times 2) = .0125$. The calibration uses the t reference distribution with the model's 27 residual degrees of freedom (Student, 1908), giving the critical value

$$t_{1-.05/(2 \times 2), 27} = 2.3734.$$

The calculations retain the same estimates and SEs but multiply each SE by this larger critical value. The following code extracts the common variance estimate from the fitted object and calculates both results from unrounded inputs.

```

mse <- deviance(one_fit) / df.residual(one_fit)
means <- with(sound_data, tapply(score, sound, mean))
estimate <- c(Instrumental = unname(means[2] - means[1]),
  Lyrics = unname(means[3] - means[1]))
se <- sqrt(mse * (1/10 + 1/10))
critical <- qt(1 - .05/(2*2), df.residual(one_fit))
raw_p <- 2 * pt(-abs(estimate/se), df.residual(one_fit))
cbind(estimate, SE = se, lower = estimate-critical*se,
  upper = estimate+critical*se,
  adjusted_p = p.adjust(raw_p, method = "bonferroni"))
#>           estimate           SE           lower           upper
#> Instrumental      -2 1.148267616 -4.725318112  0.725318116
#> Lyrics           -4 1.148267616 -6.725318112 -1.274681884
#>           adjusted_p
#> Instrumental 0.185869921135
#> Lyrics      0.003408398659

```

Table 3

Bonferroni results for the declared family of two comparisons.

Comparison	Estimate	SE	Simultaneous interval	Adjusted p
Instrumental music minus Silence	-2	1.1483	[-4.7253, 0.7253]	.1859
Music with lyrics minus Silence	-4	1.1483	[-6.7253, -1.2747]	.0034

Multiplying each raw p-value by two and capping at one gives the corresponding Bonferroni-adjusted p-value. Music with lyrics minus Silence is distinguishable from zero with this familywise calibration: its whole interval lies below zero. The Instrumental-music interval includes zero although the omnibus ANOVA rejected equality. It allows a difference from 4.7253 points below Silence to 0.7253 points above it. The two results are compatible, because the omnibus test and each contrast address different statements.

The family calibration has a purpose and a price. Chapter 4, Section 3.3, defined the familywise error rate as the probability that a declared family produces at least one false rejection when every null in it holds. Two independent comparisons, each calibrated at .05, would leave that rate at $1 - .95^2 = .0975$. These two comparisons share the Silence mean, and dividing .05 across the family bounds the rate at .05 under any dependence between them. The guarantee is paid for in width, which Section 2.3 compares, and in power. Suppose a replication of this study used three balanced groups and took this study's error SD, $\sqrt{178/27} = 2.5676$, as its planning value. A 2-point difference would first reach .8 power at 27 students per group when

the comparison stands alone, and at 33 per group when it belongs to the two-comparison family. Section 6 shows how such targets are calculated, there with the smaller planning SD of the feedback study. Declare the family from the question before seeing the results; enlarging it afterwards spends that power on questions nobody asked.

2.3 Compare with an all-pairs family

If the question includes every pair, there are three comparisons. Tukey's procedure uses the joint distribution of pairwise mean differences under the common-variance Normal-error model. `TukeyHSD()` supplies those simultaneous intervals and their adjusted p-values (Tukey, 1953).

```
TukeyHSD(one_fit, "sound", conf.level = .95)
#> Tukey multiple comparisons of means
#> 95% family-wise confidence level
#>
#> Fit: aov(formula = score ~ sound, data = sound_data)
#>
#> $sound
#>
#> diff lwr
#> Instrumental music-Silence -2 -4.84703504
#> Music with lyrics-Silence -4 -6.84703504
#> Music with lyrics-Instrumental music -2 -4.84703504
#>
#> upr p adj
#> Instrumental music-Silence 0.8470350401 0.2084569155
#> Music with lyrics-Silence -1.1529649599 0.0046975772
#> Music with lyrics-Instrumental music 0.8470350401 0.2084569155
```

The output labels each subtraction direction. Music with lyrics minus Silence is -4 , with interval $[-6.847, -1.153]$ and adjusted p .0047. Instrumental music minus Silence and Music with lyrics minus Instrumental music both equal -2 , with interval $[-4.847, 0.847]$ and adjusted p .2085. Table 4 sets one comparison, Instrumental music minus Silence, under the three calibrations met so far.

Table 4

Instrumental music minus Silence, -2 points with SE 1.1483, under three calibrations.

Calibration	Family protected	Multiplier	95% interval
Pointwise t	This comparison alone	2.0518	$[-4.3561, 0.3561]$
Bonferroni	The two comparisons with Silence	2.3734	$[-4.7253, 0.7253]$
Tukey	All three pairs	2.4794	$[-4.8470, 0.8470]$

The estimate, -2 , and its SE, 1.1483 , are the same in every row; only the multiplier changes. A larger protected family needs a larger multiplier to hold its familywise error rate at $.05$, so its intervals are wider. An interval should therefore be read together with the comparison and the family it represents.

The all-pairs family also covers a comparison chosen after seeing the means. Suppose Music with lyrics minus Silence were reported because it is the largest observed difference. That choice looked at all three pairs, so its calibration would come from the all-pairs family, $[-6.8470, -1.1530]$, rather than from the declared two-comparison family, $[-6.7253, -1.2747]$. An omnibus rejection does not by itself protect any comparison selected after examining the means. Conversely, a separately specified contrast can have its own valid uncertainty procedure without requiring omnibus rejection as an automatic preliminary gate. The relevant information is the question, selection process and calibration, rather than a universal sequence of significance decisions.

2.4 Check what changes when the variance model changes

The common-variance model assumes that within-group variability estimates one shared error variance. In a real study, design information and observed patterns help assess that working model. A small normality or variance test cannot certify it, and ten students per condition give limited information about the error distribution. In these invented data the three SDs, 2.7080 , 2.4037 and 2.5820 , are similar by construction, and every condition's residuals lie between -4 and 4 . In Figure 1 each residual is a point's vertical distance from its diamond, and the three conditions spread alike.

For independent groups whose error variances may differ, Welch's one-way procedure provides an alternative (Welch, 1947). It is the several-group counterpart of Chapter 3's Welch t test, as Chapter 4's conditions section explains. R selects it by default in `oneway.test()` (R Core Team, 2025):

```
oneway.test(score ~ sound, data = sound_data)
#>
#> One-way analysis of means (not assuming equal variances)
#>
#> data:  score and sound
#> F = 5.5144397, num df = 2.000000, denom df = 17.956056,
#> p-value = 0.01358053
```

It gives $F(2, 17.9561) = 5.5144$ and $p .0136$, beside the pooled $F(2, 27) = 6.0674$ and $p .0067$. Welch weights each condition mean by n_j/s_j^2 , the inverse of its own estimated sampling variance, and refers the statistic to fewer denominator degrees of freedom because each condition now estimates its own variance. With similar observed SDs the two procedures reach the same

conclusion at .05, while their p-values differ. Each p-value belongs to its own variance model and small-sample calibration, so the two are results to explain through those assumptions rather than votes on which numerical conclusion to prefer. With small groups or clearly unequal spreads, the model choice can matter appreciably.

A useful exact identity connects Chapter 3's equal-variance (pooled) two-sample t test, requested with `var.equal = TRUE`, with a two-group ANOVA: $F = t^2$, with the same two-sided p-value. It requires the same records, equal-variance denominator and group comparison. R's default `t.test()` is Welch's test, Chapter 3's last t procedure, and does not meet that specification. Nor is our three-group contrast the same calculation, because it uses residual information from the third condition. The Computer Lab reproduces the matched identity for Silence and Instrumental music.

3 Compare a two-way pattern

3.1 Read cells and margins before fitting

Chapter 4's second study asks how one comparison changes across a second factor. In this invented study, each of 24 pupils was individually assigned to one combination of feedback (Rereading, Answer feedback or Explanatory feedback) and practice (Massed or Spaced). The outcome is the number correct on a 20-item assessment. A cell is one combination of feedback and practice, and four pupils contribute scores to each cell in Table 5. The question is how the Spaced-minus-Massed comparison changes across feedback conditions.

Table 5

The same 24 authored scores as Chapter 4.

Feedback	Massed scores	Spaced scores
Rereading	4, 5, 7, 8	4, 5, 7, 8
Answer feedback	6, 7, 9, 10	10, 11, 13, 14
Explanatory feedback	8, 9, 11, 12	16, 17, 19, 20

These values were constructed to make the calculations transparent. They are teaching inputs, not empirical evidence that one feedback condition works better. Each score comes from a different pupil; the four scores within a cell are not repeated measurements of one pupil. Repeated patterns in their deviations were deliberately chosen and cannot establish assumptions about a population.

The following code creates the complete data. The factor levels specify the display order and the reference levels for later model coefficients.

```
feedback_data <- data.frame(
  student = 1:24,
  feedback = factor(rep(c("Rereading", "Answer feedback",
    "Explanatory feedback"), each = 8),
  levels = c("Rereading", "Answer feedback",
    "Explanatory feedback")),
  practice = factor(rep(rep(c("Massed", "Spaced"),
    each = 4), 3), levels = c("Massed", "Spaced")),
  score = c(4,5,7,8, 4,5,7,8, 6,7,9,10, 10,11,13,14,
    8,9,11,12, 16,17,19,20))
```

Computing cell counts and means before fitting any model makes the six cells and their denominators visible.

```
with(feedback_data, table(feedback, practice))
#>               practice
#> feedback      Massed Spaced
#> Rereading           4      4
#> Answer feedback     4      4
#> Explanatory feedback 4      4
cell_mean <- with(feedback_data,
  tapply(score, list(feedback, practice), mean))
cell_mean
#>               Massed Spaced
#> Rereading           6      6
#> Answer feedback     8     12
#> Explanatory feedback 10     18
```

Table 6 adds the margins and the three practice changes to these cell means.

Table 6

Cell means and equally weighted margins of the balanced design.

Feedback	Massed	Spaced	Equally weighted margin	Spaced minus Massed
Rereading	6	6	6	0
Answer feedback	8	12	10	4
Explanatory feedback	10	18	14	8
Practice margin	8	12	10	4

Each cell has four pupils, so averaging the two practice cells equally gives the feedback group's overall mean. Similarly, each practice margin averages equally across the three feedback

conditions. In an unbalanced design, a margin must specify its weights: the observed mix of pupils and an equally weighted target can answer different questions.

Figure 2 draws the six cell means with one line for each feedback condition. Each line joins two separate cell means; it does not trace individual pupils across repeated measurements.

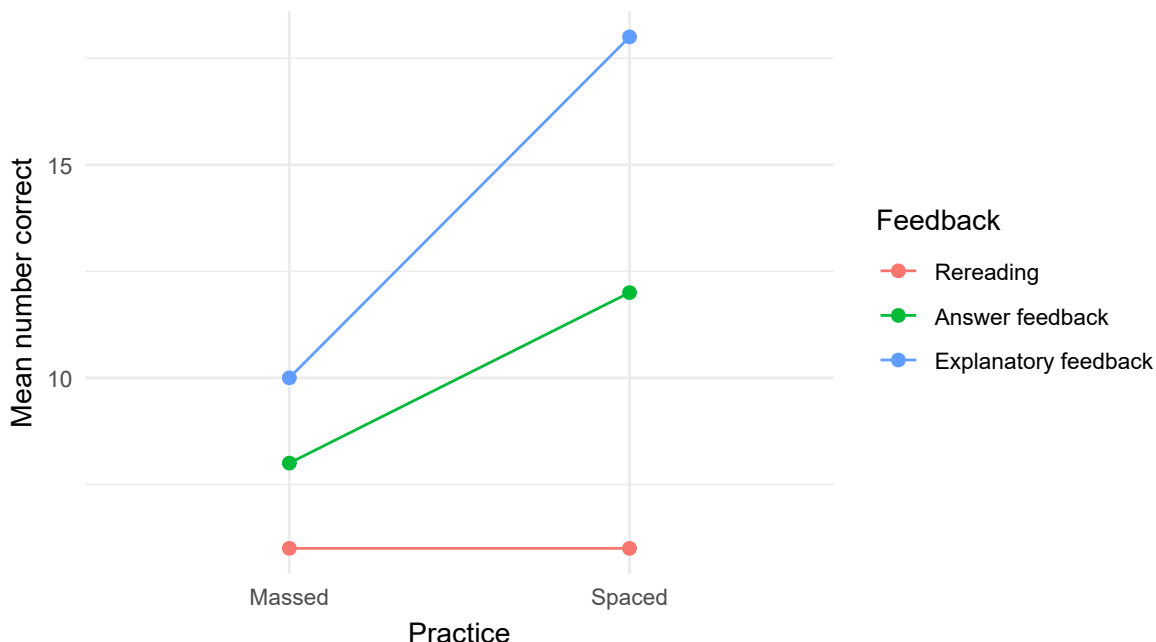


Figure 2: The practice difference depends on feedback.

The observed changes are 0, 4 and 8 points, so the three lines have different slopes. This describes interaction in the observed means. Inference about a process interaction additionally requires a model and a measure of residual variation.

3.2 Fit the full model, identify its tests and compare nested models

Chapter 4 wrote the two-way model as

$$Y_{ijk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk} + \varepsilon_{ijk},$$

where j indexes feedback, k indexes practice and i indexes pupils within a cell. With the usual sum-to-zero constraints, μ is the grand mean, α_j and β_k describe departures of the margins from it, and $(\alpha\beta)_{jk}$ describes the remaining cell departure. The constraints make this representation identifiable: the effects cannot all shift freely while leaving the same fitted means.

There is one grand-mean parameter, $3 - 1 = 2$ feedback degrees of freedom, $2 - 1 = 1$ practice degree of freedom, and $(3 - 1)(2 - 1) = 2$ interaction degrees of freedom. Together these describe six fitted cell means. The model therefore has $24 - 6 = 18$ residual degrees of freedom. In an R formula, `feedback * practice` includes both main effects and their interaction. It expands to `feedback + practice + feedback:practice`.

```
two_fit <- aov(score ~ feedback * practice,
  data = feedback_data)
anova(two_fit)
#> Analysis of Variance Table
#>
#> Response: score
#>
#>           Df Sum Sq   Mean Sq F value    Pr(>F)
#> feedback      2    256 128.000000    38.4 3.2076e-07
#> practice      1     96  96.000000    28.8 4.2288e-05
#> feedback:practice 2     64  32.000000     9.6  0.001454
#> Residuals    18     60   3.333333
```

Table 7

Full balanced factorial decomposition.

Source	SS	df	MS	F	p
Feedback	256	2	128.0000	38.4	< .0001
Practice	96	1	96.0000	28.8	< .0001
Interaction	64	2	32.0000	9.6	.0015
Residual	60	18	3.3333		
Total	476	23			

The feedback test concerns equality of feedback margins averaged equally over practice. The practice test concerns equality of practice margins averaged equally over feedback. The interaction test asks whether the practice difference is the same across all feedback conditions. With independent Normal errors of common variance, its null reference is $F_{2,18}$; the observed value 9.6 and p .0015 provide evidence against that common-difference null.

The balanced layout makes the source components orthogonal: the variation attributed to one component does not compete with another component for the same fitted direction. A single-model `anova()` table ordinarily uses sequential sums of squares, entering terms in formula order. This convenient balanced decomposition should not be extended automatically to unbalanced data.

To see why, suppose the two Explanatory–Spaced pupils who scored 19 and 20 were unavailable. The cell counts become 4, 4, 4, 4, 4 and 2, and the full model leaves $22 - 6 = 16$ residual

degrees of freedom. The equally weighted Spaced margin is then $(6 + 12 + 16.5)/3 = 11.5$, whereas the mean of the ten remaining Spaced pupils is 10.5: two answers to two different questions. Table 8 refits the full model to the 22 remaining pupils with the formula written in each order.

Table 8

Sequential sums of squares for the 22 remaining pupils, by the order of the formula.

Term	Formula <code>feedback * practice</code>	Formula <code>practice * feedback</code>
Feedback	139.7576	157.7500
Practice	52.0833	34.0909
Interaction	36.2500	36.2500
Residual	50.5000	50.5000

Sequential sums of squares give each term only what the terms entered before it have not taken. The feedback and practice rows therefore move with the order, while the interaction and the residual stay where they are. With unequal cells, one sequential table does not supply order-free tests of the main effects. Naming the two models to be compared does.

The interaction row can itself be read as a comparison of two fits to the same 24 pupils. The **additive model**, `score ~ feedback + practice`, gives every feedback condition the same Spaced-minus-Massed difference; the **full model**, `score ~ feedback * practice`, lets that difference vary. For nested models fitted to the same response and records, with residual sum of squares RSS_0 on df_0 degrees of freedom for the smaller model and RSS_1 on df_1 for the larger,

$$F = \frac{(RSS_0 - RSS_1)/(df_0 - df_1)}{RSS_1/df_1} = \frac{(124 - 60)/(20 - 18)}{60/18} = 9.6,$$

with the denominator taken from the larger model. The following code fits both models with `lm()` and compares them.

```
additive_fit <- lm(score ~ feedback + practice, data = feedback_data)
full_fit <- lm(score ~ feedback * practice, data = feedback_data)
coef(additive_fit)
#>              (Intercept)      feedbackAnswer feedback
#>                   4                   4
#> feedbackExplanatory feedback      practiceSpaced
#>                   8                   4
anova(additive_fit, full_fit)
#> Analysis of Variance Table
#>
```

```
#> Model 1: score ~ feedback + practice
#> Model 2: score ~ feedback * practice
#>   Res.Df RSS Df Sum of Sq   F    Pr(>F)
#> 1      20 124
#> 2      18  60  2      64 9.6 0.001454
```

The additive coefficients, 4 for Rereading under Massed practice, 4 and 8 for Answer and Explanatory feedback minus Rereading, and 4 for Spaced minus Massed, show the single practice difference that the smaller model imposes. The comparison returns $F(2, 18) = 9.6$ and $p = .0015$: the interaction row of Table 7, now read as a question about two named models. Because it names both models, no term's position in a formula enters it. For the 22 remaining pupils, the additive model (RSS 86.75 on 18 df) against the full model (RSS 50.50 on 16 df) gives $F(2, 16) = 5.7426$ and $p = .0132$, whichever order the formula uses. Keep the response and the records explicitly the same, because default removal of missing values can otherwise change the comparison. Chapter 9, Section 2, uses the same comparison of named models for the analysis of covariance and its parallel-slopes check.

The feedback and practice margins remain meaningful descriptions, but the interaction clarifies their limits. For example, the 4-point overall practice difference averages a 0-point, a 4-point and an 8-point difference. A result paragraph should convey this varying pattern when the scientific question concerns how practice works across feedback conditions.

3.3 Work the selected difference of differences completely

Consider Answer feedback's practice change minus Rereading's practice change:

$$\widehat{DD} = (12 - 8) - (6 - 6) = 4.$$

The contrast has coefficients $(1, -1, -1, 1)$ in the cell order Rereading–Massed, Rereading–Spaced, Answer–Massed, Answer–Spaced. Its four cell means come from different pupils. Under the common-variance model, their variances therefore add after multiplying by the squared coefficients:

$$SE(\widehat{DD}) = \sqrt{MS_E \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \right)} = 1.8257.$$

The omnibus interaction spans two independent directions. For simultaneous confidence over all contrasts in that two-dimensional interaction space, Scheffe's method uses the multiplier

$$\sqrt{2F_{.95;2,18}} = 2.6663.$$

The complete calculation keeps the cell subtraction, residual variance and family calibration visible:

```
dd <- (cell_mean["Answer feedback", "Spaced"] -
      cell_mean["Answer feedback", "Massed"]) -
      (cell_mean["Rereading", "Spaced"] -
      cell_mean["Rereading", "Massed"])
mse_two <- deviance(two_fit) / df.residual(two_fit)
se_dd <- sqrt(mse_two * 4/4)
scheffe <- sqrt(2 * qf(.95, 2, 18))
c(estimate = dd, SE = se_dd, multiplier = scheffe)
#>      estimate      SE multiplier
#> 4.0000000000 1.825741858 2.666292237
dd + c(-1, 1) * scheffe * se_dd
#> [1] -0.8679613431  8.8679613431
```

The resulting interval is $4 \pm 2.6663(1.8257) = [-0.868, 8.868]$. It includes zero even though the interaction F test rejects its null. The F test asks whether both interaction directions are zero jointly; this interval addresses one selected direction while allowing for the entire family. Other directions can carry substantial evidence, and a simultaneous interval for this direction can remain broad. Neither result cancels the other.

Table 9 sets the selected DD beside Answer feedback's simple practice effect, which happens to share its estimate, 4.

Table 9

Three quantities from the interaction section; the Scheffé interval covers the two-dimensional interaction family.

Quantity	Estimate	SE	95% interval	Calibration
Answer's simple practice effect	4	1.2910	[1.2877, 6.7123]	Pointwise
Answer-minus-Rereading DD	4	1.8257	[0.1643, 7.8357]	Pointwise
Answer-minus-Rereading DD	4	1.8257	[-0.8680, 8.8680]	Scheffé

The simple effect compares only two cell means, so its SE is smaller. It also contains the overall practice component as well as an interaction component, so it lies outside the interaction-only family, and the Scheffé guarantee does not transfer to it. The DD's two intervals differ only in their multiplier: the pointwise interval excludes zero and the simultaneous one includes it. Choosing the narrower interval after seeing which conclusion it gives does not keep the family's guarantee. State which quantity was selected, which family was considered and which interval was used before interpreting a result.

Report the interaction's magnitude in score points first. The practice difference is 0 points under Rereading, 4 under Answer feedback and 8 under Explanatory feedback, and the selected DD is 4 points with SE 1.8257. A variation share may follow once its denominator is named. The interaction sum of squares is $64/476 = .1345$ of the total sum of squares, but $64/124 = .5161$ of the interaction and residual sums of squares together. Both shares are in-sample summaries of an authored design, and neither replaces the pattern of practice differences.

4 Analyse proportions and counts

4.1 Keep the event and denominator visible

A score model is not the appropriate starting point for every outcome. Chapter 3 introduced the invented completion table in Table 10. It contains aggregate counts for 100 records, classified by learning format and completion. These records have no linkage to either score dataset above. The counts themselves establish neither random allocation nor representative sampling.

Table 10

The same authored completion counts as Chapter 3.

Format	Completed	Not completed	Total
Online	32	18	50
In person	18	32	50
Total	50	50	100

For the Online group, the event is completion and the denominator is its 50 records. Under an independent Bernoulli model with a common completion probability p , the count of completions has a binomial distribution. The estimated probability is $\hat{p} = 32/50 = .64$. If records were clustered within dependent classes or came from repeated observations of the same people, this ordinary independent-unit uncertainty model would require modification.

Suppose the benchmark is $p_0 = .5$. The score test standardizes the difference using the variance under that null:

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/50}} = \frac{.64 - .5}{\sqrt{.5(.5)/50}} = 1.9799.$$

Squaring gives $X^2 = z^2 = 3.92$, compared with a chi-squared distribution on one degree of freedom. The procedure uses a large-sample reference; the expected numbers of successes and failures under this null are both 25.

```

one_prop <- prop.test(32, 50, p = .5, correct = FALSE)
one_prop$estimate
#>      p
#> 0.64
one_prop$statistic
#> X-squared
#>      3.92
one_prop$p.value
#> [1] 0.04771488024
one_prop$conf.int
#> [1] 0.5014101688 0.7586125035
#> attr("conf.level")
#> [1] 0.95

```

The two-sided p-value is .0477. The estimated completion probability is .64, with 95% Wilson score interval [.5014, .7586]. `correct = FALSE` explicitly selects the uncorrected score procedure; accepting R's default continuity correction would specify a different calculation. The call and the reported result should identify which procedure was used.

The score interval contains the candidate probabilities that their own two-sided score tests would retain (Wilson, 1927). Each candidate supplies the variance for its null calculation. This inversion explains why the interval is linked to the score test and is not simply $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$, the Wald interval, and why its center lies slightly closer to .5 than .64 does. The Computer Lab computes the Wilson endpoints directly. The p-value assesses the stipulated benchmark; it is not the probability that $p = .5$ is true or a measure of the completion rate's practical importance.

4.2 Compare the two groups through their full table

Now ask whether completion probability differs by format. This uses both rows and a different null: format and completion are independent in the working population model, equivalently equal completion probabilities in this two-group setting. Constructing labelled counts prevents an accidental change in the event or group direction.

```

counts <- matrix(c(32, 18, 18, 32), nrow = 2,
  byrow = TRUE,
  dimnames = list(Format = c("Online", "In person"),
    Result = c("Completed", "Not completed")))
prop.table(counts, margin = 1)
#>           Result
#> Format      Completed Not completed
#>   Online           0.64           0.36
#> In person           0.36           0.64
count_test <- chisq.test(counts, correct = FALSE)
count_test$expected
#>           Result
#> Format      Completed Not completed
#>   Online           25           25
#> In person           25           25
count_test$statistic
#> X-squared
#>           7.84
count_test$p.value
#> [1] 0.005110260661
count_test$residuals
#>           Result
#> Format      Completed Not completed
#>   Online           1.4           -1.4
#> In person          -1.4           1.4

```

`margin = 1` divides each row by its own total. Completion rates are .64 for Online and .36 for In person, a difference of .28, or 28 percentage points, in the Online-minus-In-person direction. Under independence the fitted expected count in row j , column k is

$$E_{jk} = \frac{(\text{row } j \text{ total})(\text{column } k \text{ total})}{N}.$$

All four expected counts here equal $50 \times 50/100 = 25$. Pearson's statistic accumulates the squared discrepancies relative to those expected counts:

$$X^2 = \sum_{j,k} \frac{(O_{jk} - E_{jk})^2}{E_{jk}} = 4 \left(\frac{7^2}{25} \right) = 7.84.$$

For a J -by- K table the usual independence reference has $(J-1)(K-1)$ degrees of freedom. Here $df = 1$ and $p = .0051$. The observed Pearson statistic X^2 is a number calculated from the table; χ_1^2 names its approximate reference distribution under the null model.

The Pearson residuals are $(O - E)/\sqrt{E}$: 1.4, -1.4, -1.4, 1.4 in row order. Their signs locate the pattern: more Online completions and fewer Online non-completions than the independence fit predicts, with the opposite departures for In person. Squaring and adding these residuals recovers X^2 . They are related through the table margins, so their signs do not constitute four independent cell tests.

The approximation concerns expected counts under the fitted null model, along with independent observational units and a sampling model appropriate to the table. Looking only at observed cell counts misses part of this check. Sparse expected cells may require an alternative reference or a more careful model. In the present example all expected counts are 25, supporting the ordinary chi-squared approximation under the stated unit model.

The one-proportion test asked whether Online completion probability equals a fixed .5 benchmark. The complete-table test asks whether the two format probabilities are equal; it estimates their common probability under the null from the full table. Their different p-values answer different questions. Neither test establishes that changing format would cause completion to change without relevant design information.

4.3 Choose an effect scale and keep alternatives within the analysis

A chi-square test says whether the table departs from no association, not how large the association is or on which scale. Table 11 sets four common scales side by side for the completion table, with p_1 the Online and p_0 the In-person completion proportion, and gives the question that each one answers (Agresti, 2018).

Table 11

Four effect scales for the completion table, Online versus In person.

Scale	Definition	Value	What it answers
Difference in proportions	$p_1 - p_0$	.2800	How many percentage points more often the event occurs in one group: an absolute gap
Risk ratio	p_1/p_0	1.7778	How many times as probable the event is in one group: a relative gap, which changes when the other outcome is called the event
Odds ratio	$\frac{p_1/(1-p_1)}{p_0/(1-p_0)}$	3.1605	How many times as large the odds of the event are; it stays the same when rows and columns exchange roles, so a design that samples on the outcome can still estimate it
Cramér's V	$\sqrt{\frac{X^2}{N \min(J-1, K-1)}}$	.2800	How strong the association is, from 0 to 1, for a table of any size; it carries no direction and names no cell comparison

The same counts give a gap of 28 percentage points, a risk ratio of 1.7778 and an odds ratio of 3.1605. The odds ratio lies farther from 1 than the risk ratio because completion is common: Online's completion odds are $32/18 = 1.7778$ and In person's are $18/32 = .5625$, so an odds ratio is not read as a risk ratio. Calling non-completion the event changes the risk ratio to $.36/.64 = .5625$ and the odds ratio to its reciprocal, .3164, so the event and the direction are named beside every scale. Cramér's V (Cramér, 1946) equals the difference in proportions here only because every row and column total is 50. Chapter 11 reads the odds ratio again as an exponentiated logistic coefficient.

Other procedures change one part of the analysis while keeping the same table, and each answers its own question. The two-proportion interface gives a large-sample interval for the difference in proportions, a different quantity from the one-probability Wilson interval of Section 4.1. A continuity correction changes the calibration of the Pearson statistic. Fisher's exact test (Fisher, 1935) conditions on the table margins and uses an exact reference, which matters when expected counts are small. The Computer Lab runs each alternative on this table and then continues the observational student intention-by-pass example of Chapter 2, Section 3.1, where these scales, the choice of procedure and an unresolved source-coding question meet in an existing dataset.

5 Fit by an objective function and likelihood

5.1 Fix the data and vary a candidate parameter

The fitted means in Section 1 were chosen by a rule: minimize the sum of squared deviations from the observations. We can inspect this rule with the ten Silence scores, 8, 9, 9, 11, 12, 13, 13, 14, 15 and 16. These observations stay fixed. What changes is the proposed center, denoted by c .

An **objective function** assigns a numerical criterion to each candidate parameter value for fixed data. For a center fitted by least squares, the objective is

$$Q(c) = \sum_{i=1}^{10} (y_i - c)^2 = (8 - c)^2 + (9 - c)^2 + \dots + (16 - c)^2.$$

At $c = 12$, the deviations are the Silence residuals of Section 1, and their squares sum to 66 squared score points. At either 11 or 13, they sum to 76. Chapter 2, Section 2.2, states that the sample mean minimizes this criterion, and Chapter 3's Appendix A derives the identity behind it. Here the mean is 12 points. A numerical search lets us reproduce the minimizing value without writing it into the fitting instruction.

```

silence <- subset(sound_data, sound == "Silence")$score
loss <- function(center) sum((silence - center)^2)
mean_fit <- optimize(loss, interval = range(silence))
c(center = mean_fit$minimum, squared_loss = mean_fit$objective)
#>      center squared_loss
#>         12          66

```

`subset()` keeps the Section 1 rows whose condition is Silence, and `loss()` returns the sum of squared deviations for any candidate center. `optimize()` searches for a minimum of that function over the interval supplied, here `range(silence)`, or $[8, 16]$, which contains the sample mean. The objective is a single smooth bowl over this interval, so the scalar search has one minimum to locate.

The two returned quantities have different meanings. `minimum` is the **parameter location**, 12 points; `objective` is the function value there, 66 squared points. The latter is a sum of squared deviations, Silence's contribution to SS_W in Section 1, rather than an SE or an estimate of the mean. The search interval also belongs to the fitting instruction: excluding the best center would restrict which answer the search could return. Numerical stopping tolerance concerns the accuracy of that search; statistical uncertainty concerns what an estimate would do across the repetitions justified by the study.

These least-squares identities are arithmetic. An uncertainty statement for the fitted mean adds a model or design and its conditions. Chapter 7 extends this criterion from one center to a line with an intercept and slope. The Computer Lab evaluates the criterion at several centers and narrows the search interval while keeping the observations fixed.

5.2 Change the fitting criterion for a binomial outcome

For the completion example, the observations are 32 completions in 50 Online trials. Let p denote a candidate probability of completion. Under the stipulated model of independent trials with a common probability, the number completed follows a binomial distribution. The model assigns probability

$$P_p(X = 32) = \binom{50}{32} p^{32} (1 - p)^{18}.$$

In a probability calculation, we choose a value of p and consider which counts might occur. For fitting, we have already observed the count: 32 stays fixed, as does the total of 50. We now evaluate the same expression at different candidate values of p . Viewed in this direction, it is the **likelihood function**,

$$L(p) = \binom{50}{32} p^{32} (1-p)^{18}, \quad 0 \leq p \leq 1.$$

The binomial coefficient counts the ways 32 completions can occur among 50 trials. It is constant as p varies. In this binomial model, likelihood is the probability of the observed data under each candidate model parameter; it is not a probability distribution over that parameter. In particular, a high likelihood at a candidate p does not give the probability that p is true.

Maximum likelihood estimation chooses the parameter value that makes the observed data most likely under the declared model. The logarithm is increasing, so maximizing the log-likelihood gives the same parameter value. For $0 < p < 1$,

$$\ell(p) = \log \binom{50}{32} + 32 \log p + 18 \log(1-p).$$

Logs turn a product of powers into a sum and make calculation easier. The binomial coefficient affects the reported function value, although it cannot change the maximizing p . Here the interior maximum can also be found analytically. Differentiating with respect to the candidate probability gives

$$\ell'(p) = \frac{32}{p} - \frac{18}{1-p} = \frac{32 - 50p}{p(1-p)}.$$

Setting this slope to zero gives $32 - 50p = 0$, and therefore $\hat{p} = 32/50 = .64$. The denominator is positive inside $(0, 1)$. The slope is positive below .64 and negative above .64: the curve rises to this point and then falls. This verifies a maximum rather than merely finding a flat point.

To use the default minimization operation in `optimize()`, supply the **negative** log-likelihood. A smaller negative log-likelihood means a larger log-likelihood, so its minimum locates the same fitted probability.

```
negative_loglik <- function(p) {
  -dbinom(32, size = 50, prob = p, log = TRUE)
}
likelihood_fit <- optimize(negative_loglik, interval = c(0, 1))
c(numerical = likelihood_fit$minimum, analytic = 32/50,
  negative_loglik = likelihood_fit$objective)
#>      numerical      analytic negative_loglik
#>    0.6400059479    0.6400000000    2.1465474559
negative_loglik(.5)
#> [1] 4.13299674
```

`dbinom()` evaluates the binomial probability for the fixed count 32. With `log = TRUE`, it computes the log probability directly. The numerical and analytic estimates both round to .64. The small numerical difference from exactly .64 reflects the search tolerance. At the fitted probability the negative log-likelihood is 2.1465, compared with 4.1330 at $p = .5$. The smaller value is the better fit according to this criterion. The parameter estimate .64 is a completion probability; the criterion value 2.1465 is not.

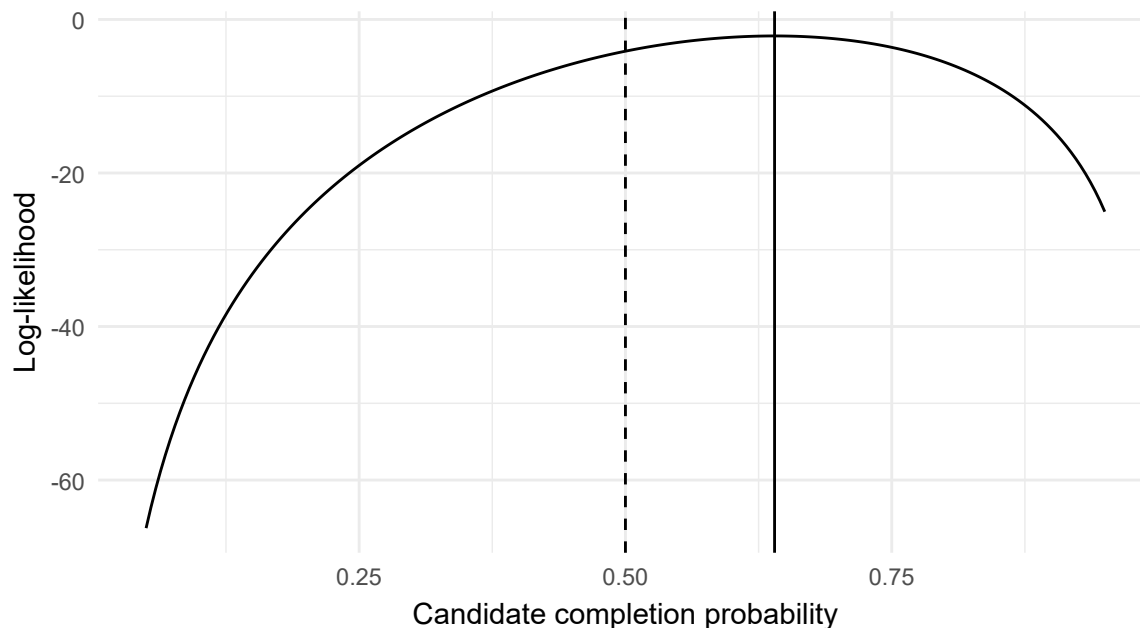


Figure 3: The binomial log-likelihood is largest at the observed proportion.

The horizontal axis contains candidate parameter values, and the vertical axis gives the log-likelihood of the same observed count at each candidate. Thus the curve displays a fitting criterion, not a sampling distribution or a distribution of probabilities over p . The peak identifies the fitted value. The score test and interval in Section 4 answer additional questions about a proposed null and the uncertainty of estimation under their stated calibration.

Finally, check the boundaries of the parameter space. With both completions and non-completions observed, $p = 0$ and $p = 1$ each give likelihood zero, so the maximum is interior. If all 50 trials instead had zero completions, the maximum would be at $p = 0$; if all completed, it would be at $p = 1$. An interior numerical search can approach such an endpoint without returning it exactly. The endpoint values must therefore be examined explicitly in those cases. Chapter 11 develops these boundary questions further when likelihood is used to fit logistic regression.

6 Examine procedures by repetition and plan precision

The preceding analyses started with observed data and produced estimates, intervals and tests. We can also begin with a declared generating model, create data from it, and examine what an analysis does over repeated studies. Chapter 5 used this approach to examine sampling variation and interval coverage. Here the questions concern an omnibus ANOVA procedure and the sample size for a specified comparison.

6.1 Follow one generated study before repeating it

Imagine three independent groups whose outcomes follow Normal distributions, each with standard deviation $\sqrt{10/3} = 1.8257$ points. First set all three population means to 8, so that the equal-means null hypothesis is true; later the unequal means 6, 8 and 10 describe an alternative. These values borrow the scale of the feedback study in Section 3, whose Massed cells have means 6, 8 and 10 with four pupils each and whose factorial fit has residual mean square 10/3. They define a hypothetical world in which we know the truth; the constructed scores do not establish that this generating model describes real pupils.

Begin with four observations per group and analyse one generated study.

```
set.seed(60630)
n_per_group <- 4
sim_group <- factor(rep(1:3, each = n_per_group))
sim_score <- rnorm(3*n_per_group,
  mean = rep(c(8, 8, 8), each = n_per_group), sd = sqrt(10/3))
anova(lm(sim_score ~ sim_group))
#> Analysis of Variance Table
#>
#> Response: sim_score
#>           Df    Sum Sq   Mean Sq F value   Pr(>F)
#> sim_group  2    7.597716  3.7988578  1.03652  0.39345
#> Residuals  9   32.985075  3.6650083
```

`rep()` and `factor()` label four observations in each of three groups, `rnorm()` draws the twelve outcomes from the declared means and common SD, and `set.seed()` makes this particular draw reproducible. The formula `sim_score ~ sim_group` fits a separate mean for each group; `anova()` compares that fit with a common mean. Its table has 2 group degrees of freedom and $12 - 3 = 9$ error degrees of freedom. For these generated data, $SS_B = 7.5977$ and $SS_W = 32.9851$, giving $F = 1.0365$ and $p = .3935$. Applying the rule $p < .05$ therefore does not reject the equal-means null in this study. Although the three generating means are identical, the three sample means vary, so the between-group sum of squares is positive. The estimated error variance, $MS_W = 3.665$, also varies around the generating variance 10/3.

One generated study cannot describe a procedure’s long-run behaviour. Repeating the same calculation many times gives a collection of p-values to compare with .05. Table 12 reports 2,000 studies for each of four conditions. The null conditions use means (8, 8, 8); the alternative conditions use (6, 8, 10). Each condition retains independent Normal observations and the same common SD, and the Computer Lab reproduces the table from seed 60630.

Table 12

Operating characteristics in the declared generating models, $B = 2000$ studies per condition.

Pupils per group	Null rejection rate	Power at means		
		MCSE	6/8/10	MCSE
4	.056	0.0051	.6545	0.0106
12	.050	0.0049	.9990	0.0007

The **null rejection rate** estimates the Type I error probability: how often this .05 omnibus procedure rejects when all three population means are equal. Both estimated rates are close to .05. The **power** column estimates the probability of rejecting that same null under the specified unequal means. It rises from .6545 with four pupils per group to .9990 with 12. This is power against the particular means 6, 8 and 10, with the stated spread and independence; a smaller separation would give different power.

Each rate is itself an estimate based on finitely many repetitions. If $\hat{q}$ is a simulated rejection proportion from B independent studies, its approximate Monte Carlo standard error is

$$MCSE(\hat{q}) = \sqrt{\frac{\hat{q}(1 - \hat{q})}{B}}.$$

For the estimated power .6545, it is $\sqrt{.6545(1 - .6545)/2000} = 0.0106$, and the table’s MCSE columns apply the same formula to each rate. This value describes the numerical uncertainty from using 2,000 simulations: a fresh set of 2,000 simulated studies can give a somewhat different power estimate. The MCSE is distinct from a pupil outcome’s SD or the SE of a group mean or contrast within one study. A rate very close to one, such as .999, also deserves care: only two of the 2,000 studies failed to reject, so its tiny plug-in MCSE rests on very few non-rejections.

Increasing the number of pupils n changes the information within a study, its error degrees of freedom and its ability to detect the specified difference. Increasing the number of repetitions B makes the simulated rate more precise; it leaves the study being simulated unchanged. Simulation describes a procedure under the declared model; agreement with its expected behaviour does not establish that the model fits a particular real study.

6.2 Plan for a specified comparison and precision target

The omnibus simulation asks whether some population group means differ. A future study may instead be planned to estimate or detect one particular difference. Consider a future balanced one-factor study of the three feedback conditions, with a declared **two-comparison Bonferroni family**: Answer feedback versus Rereading, and Explanatory feedback versus Rereading. The family has the same form as Section 2's two comparisons with Silence. Keep it, with family significance level .05, even when calculating power for just one of its comparisons.

Suppose the comparison of interest has a population difference of $\delta = 2$ points. Plan equal group sizes n , independent Normal outcomes and common SD $\sigma = \sqrt{10/3}$, the square root of the factorial fit's residual mean square in Section 3. The planning SE for a difference of two group means is $\sigma\sqrt{2/n}$. All three groups contribute to the pooled error variance, so its degrees of freedom are $3(n-1)$, even though the contrast compares only two of the means.

Bonferroni assigns two-sided significance level $.05/2 = .025$ to each comparison. Dividing that amount between its two tails puts .0125 in each tail. The rejection cutoffs are therefore $-c$ and c , where

$$c = t_{1-.05/(2 \times 2), 3(n-1)} = t_{.9875, 3(n-1)}.$$

Under the zero-difference null, the test statistic has the central t distribution already used for the comparison. Under a specified nonzero difference, its distribution moves towards that difference. The **noncentral t distribution** describes this statistic when its numerator has a nonzero mean and its denominator still estimates the common spread. Its noncentrality parameter expresses the population difference in planning SE units:

$$\lambda = \frac{\delta}{\sigma\sqrt{2/n}}.$$

For this alternative, power is the area beyond the same two rejection cutoffs: $P(T < -c) + P(T > c)$ for a noncentral t variable with $3(n-1)$ degrees of freedom and noncentrality λ . These are probabilities over future studies, rather than a p-value from observed data. Both tails belong to the two-sided procedure, even when the planning difference is positive.

```

contrast_power <- function(n, delta = 2, sigma = sqrt(10/3)) {
  df <- 3*(n-1)
  critical <- qt(1 - .05/(2*2), df)
  ncp <- delta / (sigma * sqrt(2/n))
  pt(-critical, df, ncp = ncp) +
    pt(critical, df, ncp = ncp, lower.tail = FALSE)
}
round(c(n16 = contrast_power(16), n17 = contrast_power(17)), 4)
#>      n16      n17
#> 0.7793 0.8073

```

The function accepts a proposed group size and returns power. Its first `pt()` call obtains the lower-tail probability; the second obtains the upper-tail probability. At $n = 17$, the planning SE is 0.6262, the error degrees of freedom are 48, the critical value is 2.3139, and $\lambda = 3.1937$. Summing the two tail areas gives power .8073. Checking successive integer sizes gives 17 pupils per group, 51 in total, as the first size reaching .8 power; 16 gives .7793.

This is the power of **one specified comparison within the declared family**. It is neither the probability that both comparisons reject nor the power of the omnibus F test. For example, with 12 pupils per group the specified two-point comparison has power .6328, whereas the omnibus simulation in Table 12 gives estimated power .9990 against the complete pattern (6, 8, 10). The hypotheses and rejection rules differ, so these numbers answer different questions.

A precision target concerns the interval's width rather than its probability of excluding zero. Suppose the aim is a simultaneous interval whose planned half-width is no more than one point. The family and degrees of freedom stay the same, giving

$$\text{planned half-width} = t_{.9875, 3(n-1)} \sigma \sqrt{\frac{2}{n}}.$$

This calculation substitutes the planning SD σ for the error SD that the future study will estimate. The planned half-width is 1.0078 points at 34 per group and 0.9929 at 35, so 35 per group, or 105 pupils in total, is the first size meeting this target. The realised interval uses the observed $\sqrt{MS_W}$ and therefore has a random width. Choosing 35 does not guarantee that its half-width will be at most one in every study. Figure 4 shows why a power target and an interval-width target can select quite different sample sizes.

Finally, distinguish the desired number of available outcomes from the number recruited. If outcomes are expected to be available for 90% of recruits, the 17-per-group power target needs $\lceil 17/.9 \rceil = 19$ recruits per group, 57 in total, with an expected 17.1 available outcomes per group under that retention assumption. It is an average allowance, not a guarantee of 17 outcomes in every group or a correction for bias from systematically missing outcomes.

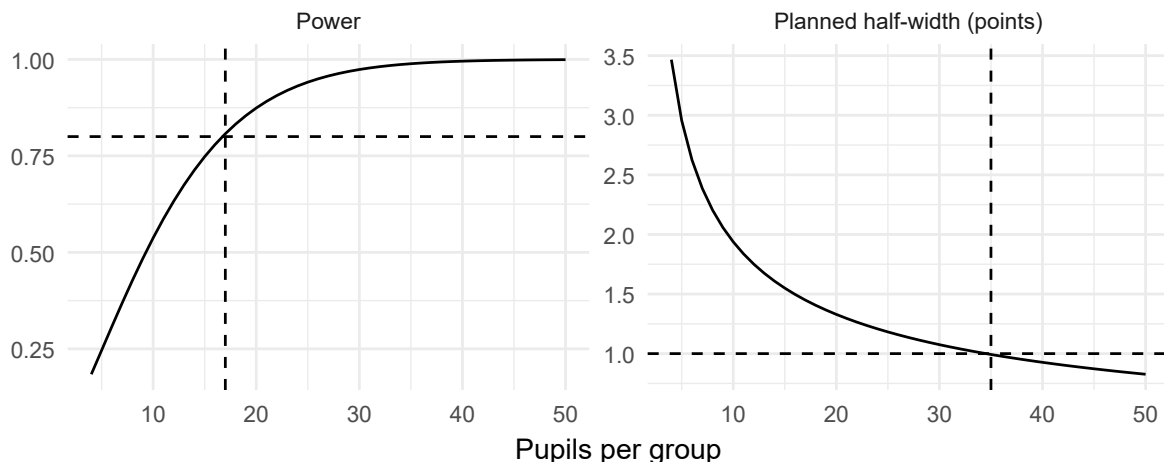


Figure 4: Power and planned interval width impose different sample-size targets.

The Computer Lab searches the full grid of sample sizes and changes the target difference and the retention rate. A two-group power calculation declares a different study, with different error degrees of freedom, and cannot silently replace this three-group calculation. Choose a planning difference from the substantive question and relevant prior evidence. As Chapter 3's significance-filter example showed, selecting statistically significant results can favour exaggerated estimates; an observed difference from such a selection is not automatically a suitable planning alternative.

7 Reading Bridge

The chapter compared group means, separated omnibus questions from calibrated contrasts, and read a comparison that changes across another factor, testing that change by comparing two named models on the same records. Counts required a different response model and their own effect scales; likelihood and an objective function made the fitting criterion explicit. Repetition examined procedures under a declared generating model, while power and precision described different planning aims.

When reading a study, identify the question, analysed records, model, estimate and uncertainty procedure before evaluating their adequacy. For a group comparison, locate its subtraction direction and the records it uses. For a test or interval, identify the null, reference distribution and comparison family. For an effect in a table, identify the event and the scale. For a simulation or planning result, locate the generating assumptions and the particular procedure being evaluated. These details connect the reported numbers to the claims they can support.

Later chapters build on each of these steps. Chapter 7 extends squared loss from a fitted center to a line: it derives the line's normal equations, relates fitted values and residuals through projection, and develops uncertainty for the slope and the fitted mean. Chapter 8 conditions on several controls and distinguishes a coefficient conditional on specified predictors from a fitted mean at a profile, a prediction for a new case and an average of fitted values over a named population. Chapter 9, Section 2, develops the analysis of covariance as regression and checks its parallel slopes with the nested-model comparison of Section 3.2. Chapter 11 develops likelihood for binary regression, including exact boundary estimates, a likelihood with more than one parameter, and the starting values and convergence checks that numerical fitting then requires.

Key terms and notation

ANOVA, contrasts, interactions and comparison families keep their main explanations in Chapter 4, and the Part One Concept Reference gives the broader reference and the principal home of each concept. The following concepts were introduced or developed through their R applications here.

Table 13

Selected concepts introduced or developed here.

Term and common alternatives	Meaning
Model formula (R formula)	Notation such as <code>score ~ group</code> naming an outcome and the variables that explain it for a fitting function. It does not state whether observations are independent or paired; that comes from the design.
Categorical effect scales (difference in proportions; risk ratio; odds ratio; Cramér's V)	Four ways to state the size of an association in a table, each answering its own question: the difference in proportions gives an absolute gap, the risk ratio a relative gap, the odds ratio a ratio of odds that stays the same when rows and columns exchange roles, and Cramér's V a unit-free strength, from 0 to 1, for a table of any size.
Objective function	A numerical criterion evaluated at candidate parameter values for fixed data; fitting seeks a stated minimum or maximum.
Likelihood and log-likelihood	The probability or density of the observed data under a declared model, viewed as a function of its parameters; the log-likelihood is its logarithm.
Maximum likelihood estimation (MLE)	Choosing parameter values that maximize the likelihood, equivalently the log-likelihood, over the stated parameter space.

The notation follows the local question. $C = \sum_j c_j \mu_j$ is a contrast, whereas $L(\theta)$ is a likelihood function of a parameter. Similarly, Type II error probability β differs from a model coefficient such as β_x . Observed Pearson X^2 differs from the reference distribution χ^2 , and $\Delta p = p_1 - p_0$ is a difference in proportions. In simulations, n counts observations per stated study or group and B counts repeated studies. An SE describes an estimator's sampling variation; an MCSE describes finite-simulation uncertainty in a computed summary.

Further reading

For the linear-model basis of ANOVA, contrasts and nested-model comparisons, see Kutner et al. (2005). Tukey (1953) develops simultaneous comparison ideas, and Wilson (1927) provides the score-interval construction for a proportion. Agresti (2018) develops the effect scales for two-way tables. Cohen (1988) discusses power and the dependence of a planning result on its chosen effect and procedure, and Çetinkaya-Rundel & Hardin (2024) develops the simulation-based treatment of operating characteristics used in Section 6. R's documentation gives the arguments and returned components for `aov`, `lm`, `TukeyHSD`, `oneway.test`, `prop.test`, `chisq.test` and `optimize` (R Core Team, 2025). The plotting chapters of Wickham et al. (2023) develop the data, mapping and layer vocabulary behind the chapter's figures, which the Computer Lab builds.

Exercises

The chapter's printed output, tables and figures supply the values each exercise reads, with a calculator for the few calculations that the reasoning needs. State the records, working conditions and comparison direction in each interpretation.

Exercise 1: Read the one-way fit from its output

For the thirty background-sound scores, `anova(one_fit)` reports sums of squares 80 and 178 on 2 and 27 df, $F = 6.0674$ and $p = .0067$, and `coef(indicator_fit)` returns 12, -2 and -4 with Silence as the reference level.

1. Say what each sum of squares measures in these scores, and why Instrumental music, whose mean equals the grand mean of 10, adds nothing to the between-group sum of squares although its ten scores vary.
2. Explain what the within-group df, 27, count, and why F divides by the residual mean square, 6.5926, rather than by the variance of all thirty scores, $258/29 = 8.966$.
3. State the equal-means null, and interpret $p = .0067$ under the working model, including what the result leaves unanswered about particular pairs of conditions.
4. Predict the three coefficients if Instrumental music were made the reference level, and say which parts of the ANOVA output would change.

Exercise 2: Keep the comparison and its family together

Instrumental music minus Silence is -2 points with SE 1.1483 on 27 df. Its 95% interval is $[-4.3561, 0.3561]$ when the comparison stands alone, $[-4.7253, 0.7253]$ inside the declared family of the two comparisons with Silence (Bonferroni) and $[-4.8470, 0.8470]$ inside the family of all three pairs (Tukey).

1. The estimate and SE are the same in all three intervals. Explain why the limits differ, naming the family that each interval protects.
2. Explain why the SE uses 27 residual df, although the comparison involves only the twenty students in two conditions.
3. An analyst reports Music with lyrics minus Silence, -4 points, because it is the largest of the three observed differences. Say which family its interval should come from, and why.
4. The omnibus F rejected equal means, yet all three intervals above include zero. Explain why these results are compatible, and why an interval that includes zero does not show that the two means are equal.

Exercise 3: Read interaction as a change in comparison

In the 24-pupil feedback-by-practice study, the Massed and Spaced cell means are 6 and 6 for Rereading, 8 and 12 for Answer feedback and 10 and 18 for Explanatory feedback, and the full two-way model gives the interaction test $F(2, 18) = 9.6$, $p = .0015$.

1. Give the three Spaced-minus-Massed changes, and say what the nonparallel lines of the interaction plot show and what they cannot show without a model.
2. The practice margins are 8 and 12. Explain why the 4-point practice main effect needs the interaction beside it when a reader asks how spaced practice works.
3. The selected difference of differences, Answer feedback's change minus Rereading's change, is 4 points with SE 1.8257. Its pointwise interval is $[0.1643, 7.8357]$, and its Scheffé interval over the two-dimensional interaction family is $[-0.8680, 8.8680]$. Explain how the Scheffé interval can include zero while the interaction F test rejects, and say which interval is warranted when this direction was chosen after the plot was inspected.
4. Answer feedback's simple practice effect is also 4 points, with SE 1.291. Explain why its SE is smaller and why the interaction family's guarantee does not cover it.

Exercise 4: Name the two models behind an interaction test

The interaction test for the 24 feedback/practice scores compares two models fitted to the same pupils: the additive model, with residual sum of squares 124 on 20 df, and the full factorial model, with 60 on 18 df.

1. Name the two models and say what the larger one adds.
2. The comparison gives $F = 9.6$ on 2 and 18 df. Explain why its denominator comes from the full model rather than the additive model.

3. After the two Explanatory-Spaced pupils scoring 19 and 20 are removed, the sequential sums of squares for feedback and practice change with the order of the formula, yet the comparison of the additive and full models gives $F(2, 16) = 5.7426$ in either order. Explain why.
4. Say what a nonsignificant result of this comparison would and would not show about additivity.

Exercise 5: Preserve the event, denominator and expected counts

In the invented table, Online has 32 completions and 18 non-completions; In person has 18 completions and 32 non-completions. For Online, the uncorrected score test of $p = .5$ gives $X^2 = 3.92$ and $p = .0477$, with the 95% Wilson interval [.5014, .7586]. For the complete table, Pearson's test gives $X^2 = 7.84$ on 1 df and $p = .0051$.

1. State the null and the denominator of each test, and explain why the two p-values answer different questions.
2. Calculate the expected count of Online completions under independence, and explain why the chi-squared approximation is checked against expected rather than observed counts.
3. The difference in proportions is .28, the risk ratio 1.7778 and the odds ratio 3.1605. Explain why the odds ratio lies farther from 1 than the risk ratio here, and why Cramér's V , .28, equals the difference in proportions only in a table like this one.
4. Say what design information a claim that the online format causes completion would additionally need.

Exercise 6: Distinguish study size, repetitions and planning targets

In the chapter's simulation, each condition repeats 2,000 generated studies of three Normal groups with common SD $\sqrt{10/3}$ and applies the .05 omnibus ANOVA. With equal means (8, 8, 8) the rejection rates are .056 at four pupils per group and .050 at twelve; with means 6, 8 and 10 the estimated power is .6545 at four and .9990 at twelve.

1. Say what the rejection rates under equal means estimate, and explain why the power rises from four to twelve pupils per group while those rates stay near .05.
2. The Monte Carlo standard error (MCSE) of the power estimate .6545 with $B = 2000$ is 0.0106. Say what it measures, and why it differs from the SE of a contrast within one simulated study.
3. Predict what quadrupling B to 8,000 would do to this MCSE and to the power that the simulation estimates.
4. For the declared two-comparison family, 17 pupils per group first reach power .8 for a 2-point comparison, and 35 per group first reach a planned simultaneous half-width of at most one point. Say what each target controls, and why a realised interval can have a different width from its planning value.

Exercise 7: Distinguish an estimate from its fitting criterion

Keep 32 successes in 50 trials fixed. Minimizing the negative binomial log-likelihood with `optimize()` over $[0, 1]$ returns `minimum` .64 and `objective` 2.1465; at $p = .5$ the negative log-likelihood is 4.1330.

1. Say which returned value is the parameter estimate and which is the criterion value, and why 2.1465 is neither a probability that $p = .64$ nor an SE.
2. The derivative of the log-likelihood is $\ell'(p) = (32 - 50p)/\{p(1 - p)\}$. Use its sign to explain why the maximum lies at 32/50.
3. Before calculating, predict how the fitted probability and the fall in log-likelihood from $p = .64$ to $p = .5$ would change with 64 successes in 100 trials. Then compare your prediction with the supplied falls, 1.9864 for 32 of 50 and 3.9729 for 64 of 100.
4. Explain why the likelihood curve is not a probability distribution over p , and what an interior numerical search would return if all 50 trials had been successes.

Exercise 8: Write a qualified result paragraph

Choose the one-way comparisons, the selected interaction contrast or the two-group count comparison from this chapter. Write a short paragraph giving the question, data and denominator, estimate, interval and its family or pointwise status, working conditions and scope. Identify one tempting claim that the numerical result does not establish, and explain the missing support.

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